

VIRSO SOFOSBUVIR TABLETS 400 mg

COMPOSITION:

Sofosbuvir Tablets 400 mg

Each tablet contains: -

Sofosbuvir IH 400 mg

DOSAGE FORM

Oral- film coated Tablets

PRODUCT DESCRIPTION

VIRSO Sofosbuvir Tablets 400mg are Blue colored, capsule shaped, biconvex bevel edged film coated tablets debossed with "400" on one side and plain on other side.

PHARMACODYNAMICS

Pharmacotherapeutic group: Direct-acting antiviral; ATC code: J05AX15.

Mechanism of action

Sofosbuvir is a pan-genotypic inhibitor of the HCV NS5B RNA-dependent RNA polymerase, which is essential for viral replication. Sofosbuvir is a nucleotide prodrug that undergoes intracellular metabolism to form the pharmacologically active uridine analog triphosphate (GS-461203), which can be incorporated into HCV RNA by the NS5B polymerase and acts as a chain terminator. In a biochemical assay, GS-461203 inhibited the polymerase activity of the recombinant NS5B from HCV genotype 1b, 2a, 3a and 4a with a 50% inhibitory concentration (IC₅₀) value ranging from 0.7 to 2.6 μ M. GS-461203 (the active metabolite of sofosbuvir) is not an inhibitor of human DNA and RNA polymerases nor an inhibitor of mitochondrial RNA polymerase.

Antiviral activity

In HCV replicon assays, the effective concentration (EC₅₀) values of sofosbuvir against full-length replicons from genotype 1a, 1b, 2a, 3a and 4a were 0.04, 0.11, 0.05, 0.05 and 0.04 μ M, respectively, and EC₅₀ values of sofosbuvir against chimeric 1b replicons encoding NS5B from genotype 2b, 5a or 6a were 0.014 to 0.015 μ M. The mean $\pm$ SD EC₅₀ of sofosbuvir against chimeric replicons encoding NS5B sequences from clinical isolates was 0.068 ± 0.024 μ M for genotype 1a (n = 67), 0.11 ± 0.029 μ M for genotype 1b (n = 29), 0.035 ± 0.018 μ M for genotype 2 (n = 15) and 0.085 ± 0.034 μ M for genotype 3a (n = 106). In these assays, the *in vitro* antiviral activity of sofosbuvir against the less common genotypes 4, 5 and 6 was similar to that observed for genotypes 1, 2 and 3.

The presence of 40% human serum had no effect on the anti-HCV activity of sofosbuvir.

Resistance

In cell culture

HCV replicons with reduced susceptibility to sofosbuvir have been selected in cell culture for multiple genotypes including 1b, 2a, 2b, 3a, 4a, 5a and 6a. Reduced susceptibility to sofosbuvir was associated with the primary NS5B substitution S282T in all replicon genotypes examined. Site-directed mutagenesis of the S282T substitution in replicons of 8 genotypes conferred 2- to 18-fold reduced susceptibility to sofosbuvir and reduced the replication viral capacity by 89% to 99% compared to the corresponding wild-type. In biochemical assays, recombinant NS5B polymerase from genotypes 1b, 2a, 3a and 4a expressing the S282T substitution showed reduced susceptibility to GS-461203 compared to respective wild-types.

In clinical studies

In a pooled analysis of 991 subjects who received sofosbuvir in Phase 3 studies, 226 subjects qualified for resistance analysis due to virologic failure or early study drug discontinuation and having HCV RNA >1,000 IU/mL. Post-baseline NS5B sequences were available for 225 of the 226 subjects, with deep sequencing data (assay cut off of 1%) from 221 of these subjects. The sofosbuvir-associated resistance substitution S282T was not detected in any of these subjects by deep sequencing or population sequencing. The S282T substitution in NS5B was detected in a single subject receiving SOFOSBUVIR monotherapy in a Phase 2 study. This subject harboured <1% HCV S282T at baseline and developed S282T (>99%) at 4 weeks post-treatment which resulted in a 13.5-fold change in sofosbuvir EC50 and reduced viral replication capacity. The S282T substitution reverted to wild-type over the next 8 weeks and was no longer detectable by deep sequencing at 12 weeks post-treatment.

Two NS5B substitutions, L159F and V321A, were detected in post-treatment relapse samples from multiple genotype 3 HCV infected subjects in the Phase 3 clinical studies. No shift in the phenotypic susceptibility to sofosbuvir or ribavirin of subject isolates with these substitutions was detected. In the study done in subjects with hepatocellular carcinoma awaiting liver transplantation where subjects received up to 48 weeks of sofosbuvir and ribavirin, the L159F substitution emerged in multiple subjects with GT1a or GT2b HCV who experienced virologic failure (breakthrough and relapse). Furthermore, the presence of substitutions L159F and/or C316N at baseline was associated with sofosbuvir breakthrough and relapse post-transplant in multiple subjects infected with GT1b HCV. In addition, S282R and L320F substitutions were detected on treatment by deep sequencing in a pre-transplant subject with GT1a HCV with a partial treatment response. The clinical significance of these findings is unknown.

Effect of baseline HCV polymorphisms on treatment outcome

Baseline NS5B sequences were obtained for 1,292 subjects from Phase 3 studies by population sequencing and the S282T substitution was not detected in any subject with available baseline sequence. In an analysis evaluating the effect of baseline polymorphisms on treatment outcome, no statistically significant association was observed between the presence of any HCV NS5B variant at baseline and treatment outcome.

Cross-resistance

HCV replicons expressing the sofosbuvir-associated resistance substitution S282T were fully susceptible to other classes of anti-HCV agents. Sofosbuvir retained activity against the NS5B substitutions L159F and L320F associated with resistance to other nucleoside inhibitors. Sofosbuvir was fully active against substitutions associated with resistance to other direct-acting antivirals with different mechanisms of actions, such as NS5B non-nucleoside inhibitors, NS3 protease inhibitors and NS5A inhibitors.

Clinical efficacy and safety

The efficacy of sofosbuvir was evaluated in five Phase 3 studies in a total of 1,568 subjects with genotypes 1 to 6 chronic hepatitis C. One study was conducted in treatment-naïve subjects with genotype 1, 4, 5 or 6 chronic hepatitis C in combination with peginterferon alfa 2a and ribavirin and the other four studies were conducted in subjects with genotype 2 or 3 chronic hepatitis C in combination with ribavirin including one in treatment-naïve subjects, one in interferon intolerant, ineligible or unwilling subjects, one in subjects previously treated with an interferon-based regimen, and one in all subjects irrespective of prior treatment history or ability to receive treatment with interferon. Subjects in these studies had compensated liver disease including cirrhosis. Sofosbuvir was administered at a dose of 400 mg once daily. The ribavirin dose was weight-based at 1,000-1,200 mg daily administered in two divided doses, and the peginterferon alfa 2a dose, where applicable, was 180 µg per week. Treatment duration was fixed in each study and was not guided by subjects' HCV RNA levels (no response guided algorithm).

Plasma HCV RNA values were measured during the clinical studies using the COBAS TaqMan HCV test (version 2.0), for use with the High Pure System. The assay had a lower limit of quantification (LLOQ) of 25 IU/mL. Sustained virologic response (SVR) was the primary endpoint to determine the HCV cure rate for all studies which was defined as HCV RNA less than LLOQ at 12 weeks after the end of treatment (SVR12).

Clinical studies in subjects with genotype 1 and 4 chronic hepatitis C

Treatment-naïve subjects - NEUTRINO (study 110)

NEUTRINO was an open-label, single-arm study that evaluated 12 weeks of treatment with sofosbuvir in combination with peginterferon alfa 2a and ribavirin in treatment-naïve subjects with genotype 1, 4, 5 or 6 HCV infection.

Treated subjects (n = 327) had a median age of 54 years (range: 19 to 70); 64% of the subjects were male; 79% were White; 17% were Black; 14% were Hispanic or Latino; mean body mass index was 29 kg/m² (range: 18 to 56 kg/m²); 78% had baseline HCV RNA greater than 6 log₁₀ IU/mL; 17% had cirrhosis; 89% had HCV genotype 1 and 11% had HCV genotype 4, 5 or 6.

Table 5 presents the response rates for the treatment group of sofosbuvir + peginterferon alfa + ribavirin.

Table 5: Response rates in study NEUTRINO

	SOF+PEG+RBV 12 weeks (n = 327)
Overall SVR12	91% (296/327)
Outcome for subjects without SVR12	
On-treatment virologic failure	0/327
Relapse ^a	9% (28/326)
Other ^b	1% (3/327)

a. The denominator for relapse is the number of subjects with HCV RNA <LLOQ at their last on-treatment assessment.

b. Other includes subjects who did not achieve SVR12 and did not meet virologic failure criteria (e.g., lost to follow-up).

Response rates for selected subgroups are presented in Table 6.

Table 6: SVR12 rates for selected subgroups in NEUTRINO

	SOF+PEG+RBV 12 weeks (n = 327)
Genotype	
Genotype 1	90% (262/292)
Genotype 1a	92% (206/225)
Genotype 1b	83% (55/66)
Genotype 4	96% (27/28)
Cirrhosis	
No	93% (253/273)
Yes	80% (43/54)
Race	
Black	87% (47/54)
Non-Black	91% (249/273)

SVR12 rates were similarly high in subjects with baseline IL28B C/C allele [94/95 (99%)] and non-C/C (C/T or T/T) allele [202/232 (87%)].

27/28 patients with genotype 4 HCV achieved SVR12. A single subject with genotype 5 and all 6 subjects with genotype 6 HCV infection in this study achieved SVR12.

Clinical studies in subjects with genotype 2 and 3 chronic hepatitis C

Treatment-naïve adults - FISSION (study 1231)

FISSION was a randomised, open-label, active-controlled study that evaluated 12 weeks of treatment with sofosbuvir and ribavirin compared to 24 weeks of treatment with peginterferon alfa 2a and ribavirin in treatment-naïve subjects with genotype 2 or 3 HCV infection. The ribavirin doses used in the sofosbuvir + ribavirin and peginterferon alfa 2a + ribavirin arms were weight-based 1,000-1,200 mg/day and 800 mg/day regardless of weight,

respectively. Subjects were randomised in a 1:1 ratio and stratified by cirrhosis (presence *versus* absence), HCV genotype (2 *versus* 3) and baseline HCV RNA level ($<6 \log_{10}$ IU/mL *versus* $\geq 6 \log_{10}$ IU/mL). Subjects with genotype 2 or 3 HCV were enrolled in an approximately 1:3 ratio.

Treated subjects (n = 499) had a median age of 50 years (range: 19 to 77); 66% of the subjects were male; 87% were White; 3% were Black; 14% were Hispanic or Latino; mean body mass index was 28 kg/m² (range: 17 to 52 kg/m²); 57% had baseline HCV RNA levels greater than 6 log₁₀ IU/mL; 20% had cirrhosis; 72% had HCV genotype 3.

Table 7 presents the response rates for the treatment groups of sofosbuvir + ribavirin and peginterferon alfa + ribavirin.

Table 7: Response rates in study FISSION

	SOF+RBV 12 weeks (n = 256)^a	PEG+RBV 24 weeks (n = 243)
Overall SVR12	67% (171/256)	67% (162/243)
Genotype 2	95% (69/73)	78% (52/67)
Genotype 3	56% (102/183)	63% (110/176)
Outcome for subjects without SVR12		
On-treatment virologic failure	< 1% (1/256)	7% (18/243)
Relapse ^b	30% (76/252)	21% (46/217)
Other ^c	3% (8/256)	7% (17/243)

a. The efficacy analysis includes 3 subjects with recombinant genotype 2/1 HCV infection.

b. The denominator for relapse is the number of subjects with HCV RNA <LLOQ at their last on-treatment assessment.

c. Other includes subjects who did not achieve SVR12 and did not meet virologic failure criteria (e.g., lost to follow-up).

The difference in the overall SVR12 rates between sofosbuvir + ribavirin and peginterferon alfa + ribavirin treatment groups was 0.3% (95% confidence interval: -7.5% to 8.0%) and the study met the predefined non-inferiority criterion.

Response rates for subjects with cirrhosis at baseline are presented in Table 8 by HCV genotype.

Table 8: SVR12 rates by cirrhosis and genotype in study FISSION

	Genotype 2		Genotype 3	
	SOF+RBV 12 weeks (n = 73)^a	PEG+RBV 24 weeks (n = 67)	SOF+RBV 12 weeks (n = 183)	PEG+RBV 24 weeks (n = 176)
Cirrhosis				
No	97% (59/61)	81% (44/54)	61% (89/145)	71% (99/139)
Yes	83% (10/12)	62% (8/13)	34% (13/38)	30% (11/37)

a. The efficacy analysis includes 3 subjects with recombinant genotype 2/1 HCV infection.

Interferon intolerant, ineligible or unwilling adults - POSITRON (study 107)

POSITRON was a randomized, double-blinded, placebo-controlled study that evaluated 12 weeks of treatment with sofosbuvir and ribavirin (n = 207) compared to placebo (n = 71) in subjects who are interferon intolerant, ineligible or unwilling. Subjects were randomized in 3:1 ratio and stratified by cirrhosis (presence versus absence).

Treated subjects (n = 278) had a median age of 54 years (range: 21 to 75); 54% of the subjects were male; 91% were White; 5% were Black; 11% were Hispanic or Latino; mean body mass index was 28 kg/m² (range: 18 to 53 kg/m²); 70% had baseline HCV RNA levels greater than 6 log₁₀ IU/mL; 16% had cirrhosis; 49% had HCV genotype 3. The proportions of subjects who were interferon intolerant, ineligible, or unwilling were 9%, 44%, and 47%, respectively. Most subjects had no prior HCV treatment (81.3%).

Table 9 presents the response rates for the treatment groups of sofosbuvir + ribavirin and placebo.

Table 9: Response rates in study POSITRON

	SOF+RBV 12 weeks (n = 207)	Placebo 12 weeks (n = 71)
Overall SVR12	78% (161/207)	0/71
Genotype 2	93% (101/109)	0/34
Genotype 3	61% (60/98)	0/37
Outcome for subjects without SVR12		
On-treatment virologic failure	0/207	97% (69/71)
Relapse ^a	20% (42/205)	0/0
Other ^b	2% (4/207)	3% (2/71)

a. The denominator for relapse is the number of subjects with HCV RNA <LLOQ at their last on-treatment assessment.

b. Other includes subjects who did not achieve SVR12 and did not meet virologic failure criteria (e.g., lost to follow-up).

The SVR12 rate in the sofosbuvir + ribavirin treatment group was statistically significant when compared to placebo ($p < 0.001$).

Table 10 presents the subgroup analysis by genotype for cirrhosis and interferon classification.

Table 10: SVR12 rates for selected subgroups by genotype in POSITRON

	SOF+RBV 12 weeks	
	Genotype 2 (n = 109)	Genotype 3 (n = 98)
Cirrhosis		
No	92% (85/92)	68% (57/84)
Yes	94% (16/17)	21% (3/14)
Interferon classification		
Ineligible	88% (36/41)	70% (33/47)
Intolerant	100% (9/9)	50% (4/8)
Unwilling	95% (56/59)	53% (23/43)

Previously treated adults - FUSION (study 108)

FUSION was a randomised, double-blinded study that evaluated 12 or 16 weeks of treatment with sofosbuvir and ribavirin in subjects who did not achieve SVR with prior interferon-based treatment (relapsers and nonresponders). Subjects were randomised in a 1:1 ratio and stratified by cirrhosis (presence *versus* absence) and HCV genotype (2 *versus* 3).

Treated subjects (n = 201) had a median age of 56 years (range: 24 to 70); 70% of the subjects were male; 87% were White; 3% were Black; 9% were Hispanic or Latino; mean body mass index was 29 kg/m² (range: 19 to 44 kg/m²); 73% had baseline HCV RNA levels greater than 6 log₁₀ IU/mL; 34% had cirrhosis; 63% had HCV genotype 3; 75% were prior relapsers.

Table 11 presents the response rates for the treatment groups of sofosbuvir + ribavirin for 12 weeks and 16 weeks.

Table 11: Response rates in study FUSION

	SOF+RBV 12 weeks (n = 103)^a	SOF+RBV 16 weeks (n = 98)^a
Overall SVR12	50% (51/103)	71% (70/98)
Genotype 2	82% (32/39)	89% (31/35)
Genotype 3	30% (19/64)	62% (39/63)
Outcome for subjects without SVR12		
On-treatment virologic failure	0/103	0/98
Relapse ^b	48% (49/103)	29% (28/98)
Other ^c	3% (3/103)	0/98

a. The efficacy analysis includes 6 subjects with recombinant genotype 2/1 HCV infection.

b. The denominator for relapse is the number of subjects with HCV RNA <LLOQ at their last on-treatment assessment.

c. Other includes subjects who did not achieve SVR12 and did not meet virologic failure criteria (e.g., lost to follow-up).

Table 12 presents the subgroup analysis by genotype for cirrhosis and response to prior HCV treatment.

Table 12: SVR12 rates for selected subgroups by genotype in study FUSION

	Genotype 2		Genotype 3	
	SOF+RBV 12 weeks (n = 39)	SOF+RBV 16 weeks (n = 35)	SOF+RBV 12 weeks (n = 64)	SOF+RBV 16 weeks (n = 63)
Cirrhosis				
No	90% (26/29)	92% (24/26)	37% (14/38)	63% (25/40)
Yes	60% (6/10)	78% (7/9)	19% (5/26)	61% (14/23)
Response to prior HCV treatment				
Relapser	86% (25/29)	89% (24/27)	31% (15/49)	65% (30/46)
Nonresponder	70% (7/10)	88% (7/8)	27% (4/15)	53% (9/17)

Treatment-naïve and previously treated adults - VALENCE (study 133)

VALENCE was a Phase 3 study that evaluated sofosbuvir in combination with weight-based ribavirin for the treatment of genotype 2 or 3 HCV infection in treatment-naïve subjects or subjects who did not achieve SVR with prior interferon-based treatment, including subjects with compensated cirrhosis. The study was designed as a direct comparison of sofosbuvir and ribavirin *versus* placebo for 12 weeks. However, based on emerging data, the study was unblinded and all HCV genotype 2 subjects continued to receive sofosbuvir and ribavirin for 12 weeks, whilst treatment for HCV genotype 3 subjects was extended to 24 weeks. Eleven HCV genotype 3 subjects had already completed treatment with sofosbuvir and ribavirin for 12 weeks at the time of the amendment.

Treated subjects (n = 419) had a median age of 51 years (range: 19 to 74); 60% of the subjects were male; median body mass index was 25 kg/m² (range: 17 to 44 kg/m²); the mean baseline HCV RNA level was 6.4 log₁₀ IU/mL; 21% had cirrhosis; 78% had HCV genotype 3; 65% were prior relapsers.

Table 13 presents the response rates for the treatment groups of sofosbuvir + ribavirin for 12 weeks and 24 weeks. Placebo recipients are not included in the tables since none achieved SVR12.

Table 13: Response rates in study VALENCE

	Genotype 2 SOF+RBV 12 weeks (n = 73)	Genotype 3 SOF+RBV 12 weeks (n = 11)	Genotype 3 SOF+RBV 24 weeks (n = 250)
Overall SVR12	93% (68/73)	27% (3/11)	84% (210/250)
Outcome for subjects without SVR12			
On-treatment virologic failure	0% (0/73)	0% (0/11)	0.4% (1/250)
Relapse ^a	7% (5/73)	55% (6/11)	14% (34/249)
Other ^b	0% (0/73)	18% (2/11)	2% (5/250)

a. The denominator for relapse is the number of subjects with HCV RNA <LLOQ at their last on-treatment assessment.

b. Other includes subjects who did not achieve SVR12 and did not meet virologic failure criteria (e.g., lost to follow-up).

Table 14 presents the subgroup analysis by genotype for cirrhosis and exposure to prior HCV treatment.

Table 14: SVR12 rates for selected subgroups by genotype in study VALENCE

	Genotype 2 SOF+RBV 12 weeks (n = 73)	Genotype 3 SOF+RBV 24 weeks (n = 250)
Treatment-naïve	97% (31/32)	93% (98/105)
Non-cirrhotic	97% (29/30)	93% (86/92)
Cirrhotic	100% (2/2)	92% (12/13)
Treatment-experienced	90% (37/41)	77% (112/145)
Non-cirrhotic	91% (30/33)	85% (85/100)
Cirrhotic	88% (7/8)	60% (27/45)

SVR12 to SVR24 concordance

The concordance between SVR12 and SVR24 (SVR 24 weeks after the end of the treatment) following treatment with sofosbuvir in combination with ribavirin or ribavirin and pegylated interferon demonstrates a positive predictive value of 99% and a negative predictive value of 99%.

Clinical efficacy and safety in special populations

HCV/HIV co-infected patients - PHOTON-1 (study 123)

Sofosbuvir was studied in an open-label clinical study evaluating the safety and efficacy of 12 or 24 weeks of treatment with sofosbuvir and ribavirin in subjects with genotype 1, 2 or 3 chronic hepatitis C co-infected with HIV-1. Genotype 2 and 3 subjects were either treatment-naïve or experienced, whereas genotype 1 subjects were naïve to prior treatment. Treatment

duration was 12 weeks in treatment-naïve subjects with genotype 2 or 3 HCV infection, and 24 weeks in treatment-experienced subjects with genotype 3 HCV infection, as well as subjects with genotype 1 HCV infection. Subjects received 400 mg sofosbuvir and weight-based ribavirin (1,000 mg for subjects weighing <75 kg or 1,200 mg for subjects weighing ≥75 kg). Subjects were either not on antiretroviral therapy with a CD4+ cell count >500 cells/mm³ or had virologically suppressed HIV-1 with a CD4+ cell count >200 cells/mm³. 95% of patients received antiretroviral therapy at the time of enrolment. Preliminary SVR12 data are available for 210 subjects.

Table 15 presents the response rates by genotype and exposure to prior HCV treatment.

Table 15: Response rates in study PHOTON-1

	Genotype 2/3 treatment-naïve SOF+RBV 12 weeks (n = 68)	Genotype 2/3 treatment experienced SOF+RBV 24 weeks (n = 28)	Genotype 1 treatment naïve SOF+RBV 24 weeks (n = 114)
Overall SVR12	75% (51/68)	93% (26/28)	76% (87/114)
Outcome for subjects without SVR12			
On-treatment virologic failure	1% (1/68)	0/28	1% (1/114)
Relapse ^a	18% (12/67)	7% (2/28)	22% (25/113)
Other ^b	6% (4/68)	0/28	1% (1/114)

a. The denominator for relapse is the number of subjects with HCV RNA <LLOQ at their last on-treatment assessment.

b. Other includes subjects who did not achieve SVR12 and did not meet virologic failure criteria (e.g., lost to follow-up).

Table 16 presents the subgroup analysis by genotype for cirrhosis.

Table 16: SVR12 rates for selected subgroups by genotype in study PHOTON-1

	HCV genotype 2		HCV genotype 3	
	SOF+RBV 12 weeks TN (n = 26)	SOF+RBV 24 weeks TE (n = 15)	SOF+RBV 12 weeks TN (n = 42)	SOF+RBV 24 weeks TE (n = 13)
Overall	88% (23/26)	93% (14/15)	67% (28/42)	92% (12/13)
No cirrhosis	88% (22/25)	92% (12/13)	67% (24/36)	100% (8/8)
Cirrhosis	100% (1/1)	100% (2/2)	67% (4/6)	80% (4/5)

TN = treatment-naïve; TE = treatment-experienced.

In subjects with HCV genotype 1 infection, the SVR rate was 82% (74/90) in subjects with genotype 1a infection and 54% (13/24) in subjects with genotype 1b infection, with relapse accounting for the majority of treatment failures. SVR rates in subjects with HCV genotype 1

infection were 80% (24/30) in subjects with baseline IL28B C/C allele and 75% (62/83) in subjects with baseline IL28B non-C/C alleles.

In the 223 CHC subjects with HIV-1 co-infection, the percentage of CD4+ cells did not change during treatment. Median CD4+ cell count decreases of 85 cells/mm³ and 84 cells/mm³ were observed at the end of treatment with sofosbuvir +ribavirin for 12 or 24 weeks, respectively. HIV-1 rebound during sofosbuvir +ribavirin treatment occurred in 2 subjects (0.9%) on antiretroviral therapy.

Patients awaiting liver transplantation - Study 2025

Sofosbuvir was studied in HCV infected subjects prior to undergoing liver transplantation in an open-label clinical study evaluating the safety and efficacy of sofosbuvir and ribavirin administered pre-transplant to prevent post-transplant HCV reinfection. The primary endpoint of the study was post-transplant virologic response (pTVR, HCV RNA <LLOQ at 12 weeks post-transplant). HCV infected subjects, regardless of genotype, with hepatocellular carcinoma (HCC) meeting the MILAN criteria received 400 mg sofosbuvir and 1,000-1,200 mg ribavirin daily for a maximum of 24 weeks, subsequently amended to 48 weeks, or until the time of liver transplantation, whichever occurred first. An interim analysis was conducted on 61 subjects who received sofosbuvir and ribavirin; the majority of subjects had HCV genotype 1, 44 subjects were CPT class A and 17 subjects were CPT class B. Of these 61 subjects, 44 subjects underwent liver transplantation following up to 48 weeks of treatment with sofosbuvir and ribavirin; 41 had HCV RNA <LLOQ at the time of transplantation.

The virologic response rates of the 41 subjects transplanted with HCV RNA <LLOQ is described in Table 17. Duration of viral suppression prior to transplantation was the most predictive factor for pTVR in those who were HCV RNA <LLOQ at the time of transplantation.

Table 17: Virologic response post-transplant in subjects with HCV RNA <LLOQ at the time of liver transplantation

	Week 12 post-transplant (pTVR)^b
Virologic response in evaluable subjects ^a	23/37 (62%)

a. Evaluable subjects are defined as those who have reached the specified time point at the time of the interim analysis.

b. pTVR: post-transplant virologic response (HCV RNA <LLOQ at 12 weeks post-procedure). In patients that discontinued therapy at 24 weeks, according to protocol, the relapse rate was 11/15.

Liver transplant recipients - Study 0126

Sofosbuvir was studied in an open label clinical study evaluating the safety and efficacy of 24 weeks of treatment with sofosbuvir and ribavirin in liver transplant recipients with chronic hepatitis C. Eligible subjects were ≥18 years old and had undergone liver transplantation 6 to 150 months prior to screening. Subjects had HCV RNA ≥10⁴ IU/mL at screening and documented evidence of chronic HCV infection pre-transplantation. The

starting dose of ribavirin was 400 mg given in a divided daily dose. If subjects-maintained haemoglobin levels ≥ 12 g/dL, ribavirin dose was increased at weeks 2, 4, and up to every 4 weeks until the appropriate weight-based dose was reached (1,000 mg daily in subjects < 75 kg, 1,200 mg daily in subjects ≥ 75 kg). The median ribavirin dose was 600 mg 800 mg daily at weeks 4-24.

Forty subjects (33 with HCV genotype 1 infection, 6 with HCV genotype 3 infection, and 1 with HCV genotype 4 infection) were enrolled, 35 of whom had previously failed interferon-based treatment, and 16 of whom had cirrhosis. 28 out of 40 (70%) subjects achieved SVR12: 22/33 (73%) with HCV genotype 1 infection, 6/6 (100%) with HCV genotype 3 infection, and 0/1 (0%) with HCV genotype 4 infection. All subjects who achieved SVR12 achieved SVR24 and SVR48.

Overview of outcomes by therapeutic regimen and treatment duration, a comparison across studies

The following tables (Table 18 to Table 21) present data from Phase 2 and Phase 3 studies relevant to the dosing to help clinicians determine the best regimen for individual patients.

Table 18: Outcomes by therapeutic regimen and treatment duration, a comparison across studies in genotype 1 HCV infection

Patient population (Study number/name)	Regimen/Duration	Subgroup	SVR12 rate % (n/N)
Treatment-naïve ^a (NEUTRINO)	SOF+PEG+RBV 12 weeks	Overall	90% (262/292)
		Genotype 1a	92% (206/225)
		Genotype 1b	83% (55/66)
		No cirrhosis	93% (253/273)
		Cirrhosis	80% (43/54)
Treatment-naïve and co-infected with HIV (PHOTON-1)	SOF+RBV 24 weeks	Overall	76% (87/114)
		Genotype 1a	82% (74/90)
		Genotype 1b	54% (13/24)
		No cirrhosis	77% (84/109)
		Cirrhosis	60% (3/5)
Treatment-naïve (QUANTUM ^b and 11-1-0258 ^b)	SOF+RBV 24 weeks	Overall ^c	65% (104/159)
		Genotype 1a ^c	69% (84/121)
		Genotype 1b ^c	53% (20/38)
		No cirrhosis ^c	68% (100/148)
		Cirrhosis ^c	36% (4/11)

n = number of subjects with SVR12 response; N = total number of subjects per group.

a. For previously treated patients with genotype 1 HCV infection, no data exists with the combination of sofosbuvir, peginterferon alfa and ribavirin. Consideration should be given to treating these patients, and potentially extending the duration of therapy with sofosbuvir, peginterferon alfa and ribavirin beyond 12 weeks and up to 24 weeks; especially for those subgroups who have one or more factors historically associated with lower response rates to interferon-based therapies (prior null response to peginterferon alfa and ribavirin therapy,

advanced fibrosis/cirrhosis, high baseline viral concentrations, black race, IL28B non CC genotype).

b. These are exploratory or Phase 2 studies. The outcomes should be interpreted with caution, as subject numbers are small and SVR rates may be impacted by the selection of patients.

c. Summary data from both studies.

Table 19: Outcomes by therapeutic regimen and treatment duration, a comparison across studies in genotype 2 HCV infection

Patient population (Study number/name)	Regimen/Duration	Subgroup	SVR12 rate % (n/N)
Treatment-naïve (FISSION)	SOF+RBV 12 weeks	Overall	95% (69/73)
		No cirrhosis	97% (59/61)
		Cirrhosis	83% (10/12)
Interferon intolerant, ineligible or unwilling (POSITRON)	SOF+RBV 12 weeks	Overall	93% (101/109)
		No cirrhosis	92% (85/92)
		Cirrhosis	94% (16/17)
Treatment-experienced (FUSION)	SOF+RBV 12 weeks	Overall	82% (32/39)
		No cirrhosis	90% (26/29)
		Cirrhosis	60% (6/10)
Treatment-naïve (VALENCE)	SOF+RBV 12 weeks	Overall	97% (31/32)
		No cirrhosis	97% (29/30)
		Cirrhosis	100% (2/2)
Treatment-experienced (VALENCE)	SOF+RBV 12 weeks	Overall	90% (37/41)
		No cirrhosis	91% (30/33)
		Cirrhosis	88% (7/8)
Treatment-experienced (FUSION)	SOF+RBV 16 weeks	Overall	89% (31/35)
		No cirrhosis	92% (24/26)
		Cirrhosis	78% (7/9)
Treatment-naïve co-infected with HIV (PHOTON-1)	SOF+RBV 12 weeks	Overall	88% (23/26)
		No cirrhosis	88% (22/25)
		Cirrhosis	100% (1/1)
Treatment-experienced co-infected with HIV (PHOTON-1)	SOF+RBV 24 weeks	Overall ^a	93% (14/15)
		No cirrhosis ^a	92% (12/13)
		Cirrhosis ^a	100% (2/2)
Treatment-naïve (ELECTRON ^b and PROTON ^b)	SOF+PEG+RBV 12 weeks	Overall ^c	96% (25/26)
Treatment-experienced (LONESTAR-2 ^b)	SOF+PEG+RBV 12 weeks	Overall	96% (22/23)
		No cirrhosis	100% (9/9)
		Cirrhosis	93% (13/14)

n = number of subjects with SVR12 response; N = total number of subjects per group.

a. These data are preliminary.

b. These are exploratory or Phase 2 studies. The outcomes should be interpreted with caution, as subject numbers are small and SVR rates may be impacted by the selection of patients. In the ELECTRON study (N = 11), the duration of peginterferon alfa ranged from 4-12 weeks in combination with sofosbuvir + ribavirin.

c. All patients were non-cirrhotic in these two studies.

Table 20: Outcomes by therapeutic regimen and treatment duration, a comparison across studies in genotype 3 HCV infection

Patient population (Study number/name)	Regimen/Duration	Subgroup	SVR12 rate % (n/N)
Treatment-naïve (FISSION)	SOF+RBV 12 weeks	Overall	56% (102/183)
		No cirrhosis	61% (89/145)
		Cirrhosis	34% (13/38)
Interferon intolerant, ineligible or unwilling (POSITRON)	SOF+RBV 12 weeks	Overall	61% (60/98)
		No cirrhosis	68% (57/84)
		Cirrhosis	21% (3/14)
Treatment- experienced (FUSION)	SOF+RBV 12 weeks	Overall	30% (19/64)
		No cirrhosis	37% (14/38)
		Cirrhosis	19% (5/26)
Treatment- experienced (FUSION)	SOF+RBV 16 weeks	Overall	62% (39/63)
		No cirrhosis	63% (25/40)
		Cirrhosis	61% (14/23)
Treatment-naïve (VALENCE)	SOF+RBV 24 weeks	Overall	93% (98/105)
		No cirrhosis	94% (86/92)
		Cirrhosis	92% (12/13)
Treatment- experienced (VALENCE)	SOF+RBV 24 weeks	Overall	77% (112/145)
		No cirrhosis	85% (85/100)
		Cirrhosis	60% (27/45)
Treatment-naïve co- infected with HIV (PHOTON-1)	SOF+RBV 12 weeks	Overall	67% (28/42)
		No cirrhosis	67% (24/36)
		Cirrhosis	67% (4/6)
Treatment- experienced co- infected with HIV (PHOTON-1)	SOF+RBV 24 weeks	Overall ^a	92% (12/13)
		No cirrhosis ^a	100% (8/8)
		Cirrhosis ^a	80% (4/5)
Treatment-naïve (ELECTRON ^b and PROTON ^b)	SOF+PEG+RBV 12 weeks	Overall ^c	97% (38/39)
Treatment- experienced (LONESTAR-2 ^b)	SOF+PEG+RBV 12 weeks	Overall	83% (20/24)
		No cirrhosis	83% (10/12)
		Cirrhosis	83% (10/12)

n = number of subjects with SVR12 response; N = total number of subjects per group.

a. These data are preliminary.

b. These are exploratory or Phase 2 studies. The outcomes should be interpreted with caution, as subject numbers are small and SVR rates may be impacted by the selection of patients. In the ELECTRON study (N = 11), the duration of peginterferon alfa ranged from 4-12 weeks in combination with sofosbuvir + ribavirin.

c. All patients were non-cirrhotic in these two studies.

Table 21: Outcomes by therapeutic regimen and treatment duration, a comparison across studies in genotype 4 HCV infection

Patient population (Study number/name)	Regimen/Duration	Subgroup	SVR12 rate % (n/N)
Treatment-naïve (NEUTRINO)	SOF + PEG + RBV 12 weeks	Overall	96% (27/28)
		No cirrhosis	100% (27/27)
		Cirrhosis	0% (0/1)

n = number of subjects with SVR12 response; N = total number of subjects per group.

Pharmacokinetics

Sofosbuvir is a nucleotide prodrug that is extensively metabolized. The active metabolite is formed in hepatocytes and not observed in plasma. The predominant (>90%) metabolite, GS-331007, is inactive. It is formed through sequential and parallel pathways to the formation of active metabolite.

Absorption

The pharmacokinetic properties of sofosbuvir and the predominant circulating metabolite GS-331007 have been evaluated. Following oral administration, sofosbuvir was absorbed quickly and the peak plasma concentration was observed, regardless of dose level. Peak plasma concentration of GS-331007 was observed between 2 to 4 hours post-dose.

Effects of food

Relative to fasting conditions, the administration of a single dose of sofosbuvir with a standardized high fat meal slowed the rate of absorption of sofosbuvir. The extent of absorption of sofosbuvir was increased approximately 1.8-fold, with little effect on peak concentration. The exposure to GS-331007 was not altered in the presence of a high-fat meal.

Distribution

Sofosbuvir is not a substrate for hepatic uptake transporters, organic anion-transporting polypeptide (OATP) 1B1 or 1B3, and organic cation transporter (OCT) 1. While subject to active tubular secretion, GS-331007 is not a substrate for renal transporters including organic anion transporter (OAT) 1 or 3, OCT2, MRP2, P-gp, BCRP or MATE1. Sofosbuvir and GS-331007 are not inhibitors of drug transporters P-gp, BCRP, MRP2, BSEP, OATP1B1, OATP1B3 and OCT1. GS-331007 is not an inhibitor of OAT1, OCT2, and MATE1.

Sofosbuvir is approximately 85% bound to human plasma proteins and the binding is independent of drug concentration over the range of 1 µg/mL to 20 µg/mL. Protein binding of GS-331007 was minimal in human plasma.

Biotransformation

Sofosbuvir is extensively metabolized in the liver to form the pharmacologically active nucleoside analog triphosphate GS-461203. The metabolic activation pathway involves sequential hydrolysis of the carboxyl ester moiety catalyzed by human cathepsin A (CatA) or carboxylesterase 1 (CES1) and phosphoramidate cleavage by histidine triad nucleotide-binding protein 1 (HINT1) followed by phosphorylation by the pyrimidine nucleotide biosynthesis pathway. Dephosphorylation results in the formation of nucleoside metabolite GS-331007 that cannot be efficiently rephosphorylated and lacks anti-HCV activity in vitro. Sofosbuvir and GS-331007 are not substrates or inhibitors of UGT1A1 or CYP3A4, CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, and CYP2D6 enzymes.

After a single 400 mg oral dose of [14C]-sofosbuvir, sofosbuvir and GS-331007 accounted for approximately 4% and >90% of drug-related material (sum of molecular weight-adjusted AUC of sofosbuvir and its metabolites) systemic exposure, respectively.

Elimination

Following a single 400 mg oral dose of [14C]-Sofosbuvir, mean total recovery of the dose was greater than 92%, consisting of approximately 80%, 14%, and 2.5% recovered in urine, faeces, and expired air, respectively. The majority of the Sofosbuvir dose recovered in urine was GS-331007 (78%) while 3.5% was recovered as sofosbuvir. This data indicate that renal clearance is the major elimination pathway for GS-331007 with a large part actively secreted. The median terminal half-lives of sofosbuvir and GS-331007 were 0.4 and 27 hours respectively.

Linearity/non-linearity

The dose linearity of sofosbuvir and its primary metabolite, GS-331007, was evaluated in fasted healthy subjects. Sofosbuvir and GS-331007 AUCs are near dose proportional over the dose range of 200 mg to 400 mg.

Pharmacokinetics in special populations

Gender and race

No clinically relevant pharmacokinetic differences due to gender or race have been identified for sofosbuvir and GS- 331007.

Elderly

Population pharmacokinetic analysis in HCV infected subjects showed that within the age range (19 to 75 years) analysed, age did not have a clinically relevant effect on the exposure to sofosbuvir and GS-331007. Clinical studies of sofosbuvir included 65 subjects aged 65 and over. The response rates observed for subjects over 65 years of age were similar to that of younger subjects across treatment groups.

Renal impairment

The pharmacokinetics of sofosbuvir were studied in HCV negative subjects with mild (eGFR ≥ 50 and < 80 mL/min/1.73 m²), moderate (eGFR ≥ 30 and < 50 mL/min/1.73 m²), severe renal impairment (eGFR < 30 mL/min/1.73 m²) and subjects with ESRD requiring haemodialysis following a single 400 mg dose of sofosbuvir. Relative to subjects with normal renal function (eGFR > 80 mL/min/1.73 m²), the sofosbuvir AUC_{0-inf} was 61%, 107% and 171% higher in mild, moderate and severe renal impairment, while the GS-331007 AUC_{0-inf} was 55%, 88% and 451% higher, respectively. In subjects with ESRD, relative to subjects with normal renal function, sofosbuvir AUC_{0-inf} was 28% higher when sofosbuvir was dosed 1 hour before haemodialysis compared with 60% higher when sofosbuvir was dosed 1 hour after haemodialysis. The AUC_{0-inf} of GS-331007 in subjects with ESRD could not be reliably determined. However, data indicate at least 10-fold and 20-fold higher exposure to GS-331007 in ESRD compared to normal subjects when SOFOSBUVIR was administered 1 hour before or 1 hour after haemodialysis, respectively.

Haemodialysis can efficiently remove (53% extraction ratio) the predominant circulating metabolite GS-331007. A 4-hour haemodialysis session removed approximately 18% of administered dose. No dose adjustment is required for patients with mild or moderate renal impairment. The safety of SOFOSBUVIR has not been assessed in patients with severe renal impairment or ESRD (to see section Warning and Precaution).

Hepatic impairment

The pharmacokinetics of sofosbuvir were studied following 7-day dosing of 400 mg sofosbuvir in HCV infected subjects with moderate and severe hepatic impairment (CPT class B and C). Relative to subjects with normal hepatic function, the sofosbuvir AUC₀₋₂₄ was 126% and 143% higher in moderate and severe hepatic impairment, while the GS-331007 AUC₀₋₂₄ was 18% and 9% higher, respectively. Population pharmacokinetics analysis in HCV infected subjects indicated that cirrhosis had no clinically relevant effect on the exposure to sofosbuvir and GS-331007. No dose adjustment of sofosbuvir is recommended for patients with mild, moderate and severe hepatic impairment (to see section Recommended Dose and Instruction for Use).

Paediatric population

The pharmacokinetics of sofosbuvir and GS-331007 in paediatric subjects have not been established (to see section Recommended Dose and Instruction for Use).

Pharmacokinetic/pharmacodynamic relationship(s)

Efficacy, in terms of rapid virologic response, has been shown to correlate with exposure to sofosbuvir as well as GS- 331007. However, neither of these entities has been evidenced to be a general surrogate marker for efficacy (SVR12) at the therapeutic 400 mg dose.

INDICATION

VIRSO is indicated in combination with other medicinal products for the treatment of chronic hepatitis C (CHC) in adults (see sections Recommended Dose, Instruction for Use, Warning and Precautions and Pharmacodynamics).

For hepatitis C virus (HCV) genotype specific activity, see sections Warning and Precautions and Pharmacodynamics.

RECOMMENDED DOSE

VIRSO treatment should be initiated and monitored by a physician experienced in the management of patients with CHC.

Posology

The recommended dose is one 400 mg tablet, taken orally, once daily with food. (see section Pharmacokinetics).

VIRSO should be used in combination with other medicinal products. Monotherapy of VIRSO is not recommended. (see section Pharmacodynamics).

Refer also to the Summary of Product Characteristics of the medicinal products that are used in combination with VIRSO. The recommended co administered medicinal product(s) and treatment duration for VIRSO combination therapy are provided in Table 1.

Table 1: Recommended co-administered medicinal product(s) and treatment duration for Sofosbuvir combination therapy

Patient population*	Treatment	Duration
Patients with genotype 1 or 4 CHC	VIRSO + ribavirin + peginterferon alfa	12 weeks ^{a,b}
Patients with genotype 2 CHC	VIRSO + ribavirin	12 weeks ^b
Patients with genotype 3 CHC	VIRSO + ribavirin + peginterferon alfa	12 weeks ^b
	VIRSO + ribavirin	24 weeks
Patients with genotype 1, 2, 3 or 4 CHC awaiting liver transplantation	VIRSO + ribavirin	Until liver transplantation ^c

*** Includes patients co-infected with human immunodeficiency virus (HIV).**

a. For previously treated patients with HCV genotype 1 infection, no data exists with the combination of Sofosbuvir, ribavirin and peginterferon alfa. (see section Warning and Precautions).

b. Consideration should be given to potentially extending the duration of therapy beyond 12 weeks and up to 24 weeks; especially for those subgroups who have one or more factors historically associated with lower response rates to interferon-based therapies (e.g. advanced fibrosis/cirrhosis, high baseline viral concentrations, black race, IL28B non-CC genotype, prior null response to peginterferon alfa and ribavirin therapy).

c. See Special patient populations – Patients awaiting liver transplantation below.

VIRSO in combination with ribavirin for 24 weeks can be considered as a therapeutic option for CHC patients with genotype 1 infection who are ineligible to receive an interferon-based regimen. (see section Warning and Precautions and Pharmacodynamics).

Treatment decision should be guided by an assessment of the potential benefits and risks for the individual patient.

The dose of ribavirin, when used in combination with VIRSO is weight based (<75 kg = 1,000 mg and ≥75 kg = 1,200 mg) and administered orally in two divided doses with food.

Concerning co administration with other direct acting antivirals against HCV, see warnings and precautions.

Dose modification

Dose reduction of VIRSO is not recommended.

If sofosbuvir is used in combination with peginterferon alfa, and a patient has a serious adverse reaction potentially related to this drug, the peginterferon alfa dose should be reduced or discontinued. Refer to the peginterferon alfa Summary of Product Characteristics for additional information about how to reduce and/or discontinue the peginterferon alfa dose.

If a patient has a serious adverse reaction potentially related to ribavirin, the ribavirin dose should be modified or discontinued, if appropriate, until the adverse reaction abates or decreases in severity. Table 2 provides guidelines for dose modifications and discontinuation based on the patient's haemoglobin concentration and cardiac status.

Table 2: Ribavirin dose modification guideline for co-administration with VIRSO

Laboratory Values	Reduce ribavirin dose to 600mg/ day if:	Discontinue ribavirin if:
Haemoglobin in subjects with no cardiac disease	<10g/dL	<8.5 g/dL
Haemoglobin in subjects with history of stable cardiac disease	≥ 2 g/dL decrease in Haemoglobin during any 4 weeks treatment period	< 12 g/dL despite 4 weeks at reduced dose

Once ribavirin has been withheld due to either a laboratory abnormality or clinical manifestation, an attempt may be made to restart ribavirin at 600 mg daily and further

increase the dose to 800 mg daily. However, it is not recommended that ribavirin be increased to the original assigned dose (1,000 mg to 1,200 mg daily).

Discontinuation of dosing

If the other medicinal products used in combination with VIRSO are permanently discontinued, VIRSO should also be discontinued (see section Warning and Precautions).

Special patient populations

Elderly

No dose adjustment is warranted for elderly patients (see section Pharmacokinetics).

Renal impairment

No dose adjustment of VIRSO is required for patients with mild or moderate renal impairment. The safety and appropriate dose of sofosbuvir have not been established in patients with severe renal impairment (estimated glomerular filtration rate [eGFR] <30 mL/min/1.73 m²) or end stage renal disease (ESRD) requiring haemodialysis (see section Pharmacokinetics).

Hepatic impairment

No dose adjustment of VIRSO is required for patients with mild, moderate or severe hepatic impairment (Child-Pugh-Turcotte [CPT] class A, B or C) (see section Pharmacokinetics).

The safety and efficacy of VIRSO have not been established in patients with decompensated cirrhosis.

Patients awaiting liver transplantation

The duration of administration of VIRSO in patients awaiting liver transplantation should be guided by an assessment of the potential benefits and risks for the individual patient (see section Pharmacodynamics).

Liver transplant recipients

VIRSO in combination with ribavirin is recommended for 24 weeks in liver transplant recipients. A starting ribavirin dose of 400 mg administered orally in two divided doses with food is recommended. If the starting dose of ribavirin is well-tolerated, the dose can be titrated up to a maximum of 1,000 - 1,200 mg daily (1,000 mg for patients weighing <75 kg and 1,200 mg for patients weighing ≥75 kg). If the starting dose of ribavirin is not well-tolerated, the dose should be reduced as clinically indicated based on haemoglobin levels (see section Pharmacodynamics).

Paediatric population

The safety and efficacy of VIRSO in children and adolescents aged <18 years have not yet been established. No data are available.

ROUTE OF ADMINISTRATION

To be administered orally

CONTRAINDICATION

Hypersensitivity to the active substance or to any of the excipients.

Medicinal products that are potent P-glycoprotein (P-gp) inducers in the intestine (rifampicin, rifabutin, St. John's wort [*Hypericum perforatum*], carbamazepine, phenobarbital and phenytoin). Co-administration will significantly decrease sofosbuvir plasma concentration and could result in loss of efficacy of VIRSO.

WARNING AND PRECAUTIONS

General

VIRSO is not recommended for administration as monotherapy and should be prescribed in combination with other medicinal products for the treatment of hepatitis C infection. If the other medicinal products used in combination with VIRSO are permanently discontinued, VIRSO should also be discontinued (see section Recommended Dose and Instruction for Use). Consult the Summary of Product Characteristics for co prescribed medicinal products before starting therapy with VIRSO.

Severe bradycardia and heart block

Cases of severe bradycardia and heart block have been observed when sofosbuvir is used in combination with another direct-acting antiviral (DAAs, including daclatasvir, simeprevir and ledipasvir) and concomitant amiodarone with or without other drugs that lower heart rate. The mechanism is not established.

The concomitant use of amiodarone was limited through the clinical development of sofosbuvir plus DAAs. Cases are potentially life threatening, therefore amiodarone should only be used in patients on VIRSO and another DAA when other alternative anti-arrhythmic treatments are not tolerated or are contraindicated. Patients also taking beta blockers, or those with underlying cardiac comorbidities and/or advanced liver disease may be at increased risk for symptomatic bradycardia with coadministration of amiodarone.

Should concomitant use of amiodarone be considered necessary it is recommended that patients are closely monitored when initiating VIRSO and another DAA. Patients who are identified as being at high risk of bradyarrhythmia should be continuously monitored for 48 hours in an appropriate clinical setting.

Due to the long half-life of amiodarone, appropriate monitoring should also be carried out for patients who have discontinued amiodarone within the past few months and are to be initiated on VIRSO in combination with and another DAA.

All patients receiving VIRSO and another DAA in combination with amiodarone with or without other drugs that lower heart rate should also be warned of the symptoms of bradycardia and heart block and should be advised to seek medical advice urgently should they experience them.

Treatment-experienced patients with genotype 1 and 4 HCV infection

Sofosbuvir has not been studied in a Phase 3 study in treatment experienced patients with genotype 1 and 4 HCV infection. The optimal treatment duration in this population has not been established (see also sections Recommended Dose, Instruction for Use and Pharmacodynamics).

Consideration should be given to treating these patients, and potentially extending the duration of therapy with sofosbuvir, peginterferon alfa and ribavirin beyond 12 weeks and up to 24 weeks; especially for those subgroups who have one or more factors historically associated with lower response rates to interferon based therapies (advanced fibrosis/cirrhosis, high baseline viral concentrations, black race, IL28B non-CC genotype).

Treatment of patients with genotype 5 or 6 HCV infection

There is very limited data regarding the use of Sofosbuvir in patients with genotype 5 and 6 HCV infection (see section Pharmacodynamics). Available data on subjects with genotype 5 or 6 HCV infection are insufficient for dosing recommendations.

Interferon-free therapy for genotype 1 HCV infection

Interferon free regimens for patients with genotype 1 HCV infection with sofosbuvir have not been investigated in Phase 3 studies (see section Pharmacodynamics). The optimal regimen and treatment duration have not been established. Such regimens should only be used for patients that are intolerant to or ineligible for interferon therapy, and are in urgent need of treatment.

Co-administration with other direct-acting antivirals against HCV

VIRSO should only be co administered with other direct acting antiviral medicinal products if the benefit is considered to outweigh the risks based upon available data. There are no data to support the co administration of sofosbuvir and telaprevir or boceprevir. Such co administration is not recommended (see also section Interactions with Other Medicaments).

Pregnancy and concomitant use with ribavirin

When VIRSO is used in combination with ribavirin or peginterferon alfa/ribavirin, women of childbearing potential or their male partners must use an effective form of contraception during the treatment and for a period of time after the treatment as recommended in the Summary of Product Characteristics for ribavirin. Refer to the Summary of Product Characteristics for ribavirin for additional information.

Use with moderate P-gp inducers

Medicinal products that are moderate P-gp inducers in the intestine (e.g. oxcarbazepine and modafinil) may decrease sofosbuvir plasma concentration leading to reduced therapeutic effect of VIRSO. Co administration of such medicinal products is not recommended with sofosbuvir (see section Interactions with Other Medicaments).

Renal impairment

The safety of sofosbuvir has not been assessed in subjects with severe renal impairment (eGFR <30 mL/min/1.73 m²) or ESRD requiring haemodialysis. Furthermore, the appropriate dose has not been established. When VIRSO is used in combination with ribavirin or peginterferon alfa/ribavirin, refer also to the Summary of Product Characteristics for ribavirin for patients with creatinine clearance (CrCl) <50 mL/min (see also section Pharmacokinetic).

HCV/HBV (hepatitis B virus) co-infection

Cases of hepatitis B virus (HBV) reactivation, some of them fatal, have been reported during or after treatment with direct-acting antiviral agents. HBV screening should be performed in all patients before initiation of treatment. HBV/HCV co-infected patients are at risk of HBV reactivation, and should therefore be monitored and managed according to current clinical guidelines.

Paediatric population

VIRSO is not recommended for use in children and adolescents under 18 years of age because the safety and efficacy have not been established in this population.

Elderly patients

Clinical studies of sofosbuvir included 65 subjects aged 65 and over. The response rates observed for subjects over 65 years of age were similar to that of younger subjects across treatment groups. In general, caution should be exercised when administering sofosbuvir in elderly patients reflecting the greater frequency of anaemia, decreased hepatic, renal or cardiac function, and of concomitant disease or other drug therapy.

Post-liver transplant patients

The safety and efficacy of VIRSO have not been established in post-liver transplant patients.

INTERACTIONS WITH OTHER MEDICAMENTS

Sofosbuvir is a nucleotide prodrug. After oral administration of Sofosbuvir, sofosbuvir is rapidly absorbed and subject to extensive first-pass hepatic and intestinal metabolism. Intracellular hydrolytic prodrug cleavage catalyzed by enzymes including carboxylesterase 1 and sequential phosphorylation steps catalyzed by nucleotide kinases result in formation of the pharmacologically active uridine nucleoside analogue triphosphate. The predominant inactive circulating metabolite GS-331007 that accounts for greater than 90% of drug-related material systemic exposure is formed through pathways sequential and parallel to formation

of active metabolite. The parent sofosbuvir accounts for approximately 4% of drug-related material systemic exposure.

Sofosbuvir is a substrate of drug transporter P-gp and breast cancer resistance protein (BCRP) while GS-331007 is not.

Medicinal products that are potent P-gp inducers in the intestine (e.g. rifampicin, St. John's wort, carbamazepine and phenytoin) may decrease sofosbuvir plasma concentration leading to reduced therapeutic effect of Sofosbuvir and thus should not be used with Sofosbuvir. Medicinal products that are moderate P-gp inducers in the intestine (e.g. oxcarbazepine and modafinil) may decrease sofosbuvir plasma concentration leading to reduced therapeutic effect of sofosbuvir. Co-administration with such medicinal products is not recommended with Sofosbuvir. Co-administration of Sofosbuvir with medicinal products that inhibit P-gp and/or BCRP may increase sofosbuvir plasma concentration without increasing GS-331007 plasma concentration, thus Sofosbuvir may be co-administered with P-gp and/or BCRP inhibitors. Sofosbuvir and GS-331007 are not inhibitors of P-gp and BCRP and thus are not expected to increase exposures of medicinal products that are substrates of these transporters.

The intracellular metabolic activation pathway of sofosbuvir is mediated by generally low affinity and high capacity hydrolase and nucleotide phosphorylation pathways that are unlikely to be affected by concomitant medicinal products.

Patients treated with vitamin K antagonists

As liver function may change during treatment with VIRSO, a close monitoring of International Normalised Ratio (INR) values is recommended.

Other interactions

Drug interaction information for Sofosbuvir with potential concomitant medicinal products is summarized in Table 3 below. The table is not all-inclusive.

Table 5: Interactions between VIRSO and other medicinal products

Medicinal product by therapeutics areas	Recommendation concerning co-administration with Sofosbuvir
ANALEPTICS	
Modafinil	Co-administration of sofosbuvir with modafinil is expected to decrease the concentration of sofosbuvir, leading to reduced therapeutic effect of sofosbuvir. Such co-administration is not recommended.
ANTIARRHYTHMICS	
Amiodarone	Use only if no other alternative is available. Close monitoring is recommended if this medicinal product is administered with VIRSO

ANTICOAGULANTS	
Vitamin K antagonists	Close monitoring of INR is recommended with all vitamin K antagonists. This is due to liver function changes during treatment with VIRSO.
ANTICONSULSANTS	
Carbamazepine Phenytoin Phenobarbital	Sofosbuvir is contraindicated with carbamazepine, phenobarbital and phenytoin, potent intestinal P-gp inducers.
Oxcarbazepine	Co-administration of VIRSO with oxcarbazepine is expected to decrease the concentration of sofosbuvir, leading to reduced therapeutic effect of VIRSO. Such co-administration is not recommended.
ANTIMYCOBACTERIALS	
Rifampicin (600mg Single Dose)	VIRSO is contraindicated with rifampicin, a potent intestinal P-gp inducer.
Rifabutin Rifapentine	VIRSO is contraindicated with rifabutin, a potent intestinal P-gp inducer. Co-administration of VIRSO with rifapentine is expected to decrease the concentration of sofosbuvir, leading to reduced therapeutic effect of VIRSO. Such co-administration is not recommended.
HERBAL SUPPLEMENTS	
St. John's Wort (Hypericum perforatum)	Sofosbuvir should not be used with St. John's Wort, potent intestinal P-gp inducer.
HCV ANTIVIRAL AGENTS: HCV PROTEASE INHIBITORS	
Boceprevir (BOC) Telaprevir (TPV)	No drug-drug interactions data exists regarding the co-administration of Sofosbuvir with boceprevir or telaprevir.
NARCOTIC ANALGESICS	
Methadone (Methadone maintenance therapy [30 to 130 mg/daily])	No dose adjustment of sofosbuvir or methadone is required when sofosbuvir and methadone are used concomitantly.
IMMUNOSUPPRESSANTS	
Ciclosporin (600mg single dose)	No dose adjustment of sofosbuvir or Ciclosporin is required when sofosbuvir and Ciclosporin are used concomitantly.
Tacrolimus (5 mg single dose)	No dose adjustment of sofosbuvir or tacrolimus is required when sofosbuvir and tacrolimus are used concomitantly.
HIV ANTIVIRAL AGENTS: REVERSE TRANSCRIPTASE INHIBITORS	
Efavirenz (600mg once daily)	No dose adjustment of sofosbuvir or Efavirenz is required when sofosbuvir and Efavirenz are used concomitantly.
Emtricitabine (200 mg once daily)	No dose adjustment of sofosbuvir or emtricitabine is required when sofosbuvir and emtricitabine are used concomitantly.

Tenofovir disoproxil fumarate (245 mg once daily)	No dose adjustment of sofosbuvir or tenofovir disoproxil is required when sofosbuvir and tenofovir disoproxil are used concomitantly.
Rilpivirine (25 mg once daily)	No dose adjustment of sofosbuvir or rilpivirine is required when sofosbuvir and rilpivirine are used concomitantly.
HIV ANTIVIRAL AGENTS: HIV PROTEASE INHIBITORS	
Darunavir boosted with ritonavir (800/100 mg once daily)	No dose adjustment of sofosbuvir or darunavir (ritonavir boosted) is required when sofosbuvir and darunavir are used concomitantly.
HIV ANTIVIRAL AGENTS: INTEGRASE INHIBITORS	
Raltegravir (400 mg twice daily)	No dose adjustment of sofosbuvir or raltegravir is required when sofosbuvir and raltegravir are used concomitantly.
ORAL CONTRACEPTIVES	
Norgestimate/ethinyl estradiol	No dose adjustment of norgestimate/ethinyl estradiol is required when sofosbuvir and norgestimate/ethinyl estradiol are used concomitantly.

PREGNANCY AND LACTATION

Women of childbearing potential / contraception in males and females

When Sofosbuvir is used in combination with ribavirin or peginterferon alfa/ribavirin, extreme care must be taken to avoid pregnancy in female patients and in female partners of male patients. Women of childbearing potential or their male partners must use an effective form of contraception during treatment and for a period of time after the treatment has concluded as recommended in the Summary of Product Characteristics for ribavirin. Refer to the Summary of Product Characteristics for ribavirin for additional information.

Pregnancy

There are no or limited amount of data from the use of sofosbuvir in pregnant women.

As a precautionary measure, it is preferable to avoid the use of Sofosbuvir during pregnancy. However, if ribavirin is co-administered with sofosbuvir, the contraindications regarding use of ribavirin during pregnancy apply (see also the Summary of Product Characteristics for ribavirin).

Breast-feeding

It is unknown whether sofosbuvir and its metabolites are excreted in human milk. A risk to new-borns/infants cannot be excluded. Therefore, Sofosbuvir should not be used during breast-feeding.

Fertility

No human data on the effect of Sofosbuvir on fertility are available.

SIDE EFFECTS

Summary of the safety profile

During treatment with sofosbuvir in combination with ribavirin or with peginterferon alfa and ribavirin, the most frequently reported adverse drug reactions were consistent with the expected safety profile of ribavirin and peginterferon alfa treatment, without increasing the frequency or severity of the expected adverse drug reactions.

Tabulated summary of adverse reactions

Sofosbuvir has mainly been studied in combination with ribavirin, with or without peginterferon alfa. In this context, no adverse drug reactions specific to sofosbuvir have been identified. The most common adverse drug reactions occurring in subjects receiving sofosbuvir and ribavirin or sofosbuvir, ribavirin and peginterferon alfa were fatigue, headache, nausea and insomnia.

The following adverse drug reactions have been identified with sofosbuvir in combination with ribavirin or in combination with peginterferon alfa and ribavirin (Table 4).

Frequency	SOF ^a + RBV ^b	SOF + PEG ^c + RBV
<i>Infections and infestations:</i>		
Common	nasopharyngitis	
<i>Blood and lymphatic system disorders:</i>		
Very common	haemoglobin decreased	anaemia, neutropenia, lymphocyte count decreased, platelet count decreased
Common	anaemia	
<i>Metabolism and nutrition disorders:</i>		
Very common		decreased appetite
Common		weight decreased
<i>Psychiatric disorders:</i>		
Very common	insomnia	insomnia
Common	depression	depression, anxiety, agitation
<i>Nervous system disorders:</i>		
Very common	headache	dizziness, headache

Common	disturbance in attention	migraine, memory impairment, disturbance in attention
<i>Eye disorders:</i>		
Common		vision blurred
<i>Respiratory, thoracic and mediastinal disorders:</i>		
Very common		dyspnoea, cough
Common	dyspnoea, dyspnoea exertional, cough	dyspnoea exertional
<i>Gastrointestinal disorders:</i>		
Very common	nausea	diarrhoea, nausea, vomiting
Common	abdominal discomfort, constipation, dyspepsia	constipation, dry mouth, gastroesophageal reflux
<i>Hepatobiliary disorders:</i>		
Very common	blood bilirubin increased	blood bilirubin increased
<i>Skin and subcutaneous tissue disorders:</i>		
Very common		rash, pruritus
Common	alopecia, dry skin, pruritus	alopecia, dry skin
<i>Musculoskeletal and connective tissue disorders:</i>		
Very common		arthralgia, myalgia
Common	arthralgia, back pain, muscle spasms, myalgia	back pain, muscle spasms
<i>General disorders and administration site conditions:</i>		
Very common	fatigue, irritability	chills, fatigue, influenza-like illness, irritability, pain, pyrexia
Common	pyrexia, asthenia	chest pain, asthenia

a. SOF = sofosbuvir; b. RBV = ribavirin; c. PEG = peginterferon alfa.

Other special population(s)

HIV/HCV co-infection

The safety profile of sofosbuvir and ribavirin in HCV/HIV co-infected subjects was similar to that observed in mono-infected HCV treated with sofosbuvir and ribavirin.

Patients awaiting liver transplantation

The safety profile of sofosbuvir and ribavirin in HCV infected subjects prior to liver transplantation was similar to that observed when treated with sofosbuvir and ribavirin.

SYMPTOMS AND TREATMENT OF OVERDOSE

The effects of higher doses are unknown.

No specific antidote is available for overdose with Sofosbuvir. If overdose occurs the patient must be monitored for evidence of toxicity. Treatment of overdose with Sofosbuvir consists of general supportive measures including monitoring of vital signs as well as observation of the clinical status of the patient. Haemodialysis can efficiently remove (53% extraction ratio) the predominant circulating metabolite GS-331007. A 4-hour haemodialysis session removed 18% of the administered dose.

EFFECT ON ABILITY TO DRIVE AND USE MACHINE

Sofosbuvir has moderate influence on the ability to drive and use machines. Patients should be informed that fatigue and disturbance in attention, dizziness and blurred vision have been reported during treatment with sofosbuvir in combination with peginterferon alfa and ribavirin.

PRECLINICAL SAFETY DATA

Not Applicable

INSTRUCTION FOR USE

The film-coated tablet is for oral use. Patients should be instructed to swallow the tablet whole. The film-coated tablet should not be chewed or crushed, due to the bitter taste of the active substance.

The tablet should be taken with food (**see section Pharmacokinetics**).

Patients should be instructed that if vomiting occurs within 2 hours of dosing an additional tablet should be taken. If vomiting occurs more than 2 hours after dosing, no further dose is needed. These recommendations are based on the absorption kinetics of sofosbuvir and GS-331007 suggesting that the majority of the dose is absorbed within 2 hours after dosing. If a dose is missed and it is within 18 hours of the normal time, patients should be instructed to take the tablet as soon as possible and then patients should take the next dose at the usual time. If it is after 18 hours then patients should be instructed to wait and take the next dose at the usual time.

Patients should be instructed not to take a double dose.

STORAGE CONDITION

Store below 30°C and protect from moisture.

SHELF LIFE

3 years.

THERAPEUTICS CODE

ATC code : J05AP08

PRESENTATIONS

28 Tablets are packed in 100cc, Round, Screw, White Opaque HDPE Container, with 38/400 Neck Child Resistant, outer-white Opaque PP Cap with induction sealing HS 130-35 liner, printed as "Sealed for your protection".

Each bottle contains desiccant – 1g Silica Gel Sachet and pharmaceutical grade polyester coil: 9g/ Yard.

Each Bottle is packed in an outer Carton along with leaflet.

MANUFACTURED BY:

STRIDES PHARMA SCIENCE LIMITED

36/7, Suragajakkanahalli,
Indlavadi cross, Anekal Taluk
Bangalore – 562 106, INDIA

PRODUCT REGISTRATION HOLDER:

UNIMED SDN BHD

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DATE OF REVISION

6th November 2019